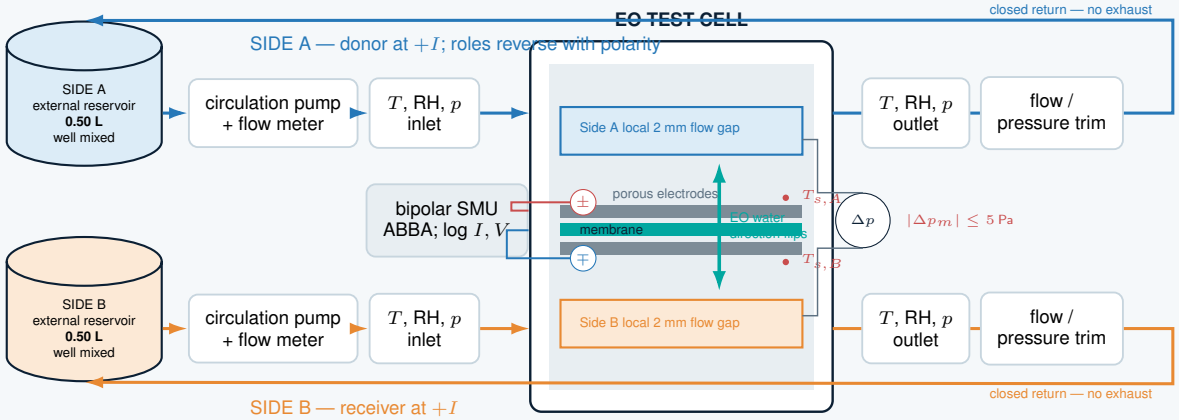
Small EO vapour coupon cell — closed-loop setup and key dimensions

Each side: 500 mL external well-mixed reservoir + 6.22 mL test cell; the 2 mm dimension is only the local membrane flow gap.

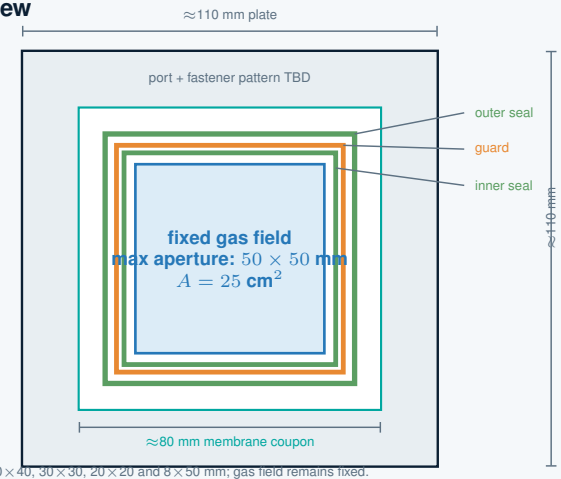
A Reversible closed-loop measurement arrangement



Initial condition: both reservoirs preconditioned at $32\,^{\circ}$C, $a_w = 0.50$, $p = 101.325$ kPa; close the measured water inventory.

The pump recirculates the same gas; it does not replace it. Reservoir volume controls RH drift, while the 2 mm gap controls local gas-film transport.

B Plan view



Mask inserts: 40×40, 30×30, 20×20 and 8×50 mm; gas field remains fixed.
Radial lands: active–coupon = 15 mm; coupon–plate = 15 mm.
Coupon land: 3 mm inner seal + 2 mm guard + 3 mm outer seal + 7 mm spare.

C Local membrane-cell cross-section (external reservoirs not shown)



The 2 mm dimension is the local membrane flow gap only — NOT the depth of the 0.50-L external reservoir.

Hard stops: 1 / 2 / 4 mm. Internal cell = 6.22 mL/side; external reservoir = 500 mL/side.

D Where the gas volume is located (one side shown)

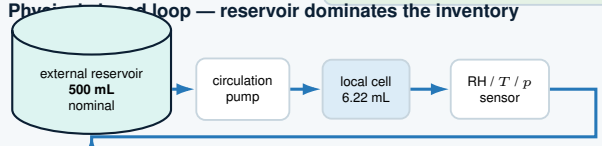
Internal cell: $V_{cell} = 6.22$ mL/side
5.00 mL active + 1.22 mL headers/ports/taps
non-active target $\leq$1.3 mL/side

External gas reservoir
$V_{res} = 500$ mL per side
size reference: cylinder ≈80 mm diameter × 100 mm; mix and thermostat

Closed gas inventory per side
$V_{gas} \approx 506.2$ mL + lines/sensor reservoir + cell + all connected voids

Recirculation scale
$V_{gas}/Q \approx 30.4$ s at 1.00 L/min
mixing/pass frequency — not fresh-gas replacement

Physical closed loop — reservoir dominates the inventory



The external reservoir contains ≈80 times more gas than the local test cell.

At 0.144 mg/s, 0.50 L still gives an ideal 10-RH-point side-to-side drift in ≈5.9 s.

Use interchangeable 0.5 / 1 / 2 / 5 L reservoirs when a slower transient is required; keep the local gap at 2 mm.

This closed-loop mode demonstrates transient pumping and polarity reversal.
Use a separate once-through test for sustained steady flux; measure every connected gas and surface-water inventory.

Design basis, not a manufacturing drawing: verify seal compression and swept volume; accept loaded bow ≤ 20 µm for the 1 mm gap and map the assembled gaps.